ALP-ttbar Toy MC v1.7: Post-registration Results for a Three-Dimensional Two-Hypothesis Vertex Fit (Adding the r–z Opening Angle) of Displaced Light ALP → e^+e^-

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Preregistration (frozen and public before any v1.7 result, including an amendment that added V-allpixel and P7 before any computation): doi:10.5281/zenodo.23995668, github.com/wspaik5/ALP_ttbar_Toy_MC_v1.7_Preregistration_3D_Vertex_Fit (README SHA-256 b538ea63...1ec3beb). Previous versions: v1.6 doi:10.5281/zenodo.23952888, v1.5 doi:10.5281/zenodo.23924862

Disclaimer. This is an independent, simplified, numpy-only toy study. It is not affiliated with the authors of Refs. [1, 2] or with the CMS, LHCb or ALICE Collaborations or the FCC study, uses parametrised kinematics and assumed detector numbers, and should be read as qualitative trends only. It makes no claim about the design or performance of a real detector.

Summary. v1.5–v1.6 used only the transverse view, where a free vertex absorbs a small ALP opening and bremsstrahlung fakes openings. v1.7 adds the r–z view (z from the pixel layers; variants with stereo strips and with z at every layer) and an r–z two-hypothesis test on the unchanged transverse fit. **Bremsstrahlung barely fakes an r–z opening** (0.84% of conversions with $\Delta\chi^2_L > 9$, against 9.6% in the transverse view), and the r–z test separates ALP pairs well from 50 MeV on. The loss in reach falls at 50–200 MeV (200 MeV: 8.2× → 6.1× pixel z, 4.5× with stereo, 3.9× with z at every layer), but **not at 17 MeV** (127× → 127×; 117× even with z at every layer). Six of seven predictions are confirmed; P7 (≥ 25% gain at 17 MeV with z at every layer) is refuted. All three validations passed, and the preregistered null references (χ^2_1 for the r–z test, $½\chi^2_1 + ½\chi^2_2$ for the 3D sum) were confirmed. One sanity-gate failure was found and fixed (§5).

1. Question and preregistration

Alimena et al. [1] extended the top-associated long-lived ALP search of Rygaard et al. [2] to $a \to e^+e^-$ at generator level; this toy series studies detector-level effects not included there. In the r–z view the tracks are straight in the transverse path length s ($z = z_v + \cot\theta \cdot s$), a vertex shift cannot absorb a polar opening, bremsstrahlung barely changes the polar angle, and conversions have $\Delta\cot\theta \approx 0$ [3]. Using the r–z opening against conversions and 3D vertex fitting are standard [6–8]; the claim is limited to quantifying, in this toy, how much the r–z opening recovers the conversion-induced loss. **Q1:** how large is the degradation at 17–200 MeV ($c\tau$ = 1 mm) with the 3D selection, compared with the transverse-only result on the same subsample (v1.6 config R)? **Q2:** how are $\Delta\chi^2_L$ and $\Delta\chi^2_{3D}$ distributed for conversions and ALP pairs?

2. Method (as preregistered, with one deviation in §5)

Unchanged: v1.6 config R (3.8 T, nominal material), the same subsample (100,000 conversion and 60,000 signal pairs per mass), hits, H_{conv} and H_{ALP} transverse fits and $\Delta\chi^2_T$; selection scan_v15 with $\Delta\chi^2_{3D} = \Delta\chi^2_T + \Delta\chi^2_L$. **Longitudinal simulation:** z along the transverse path of the transverse simulation; at every material surface after the vertex a Highland kick of the polar angle with the momentum of the simulated track (bremsstrahlung included); separate random stream (seeds 1031, 1041), so the transverse hits are identical to v1.5/v1.6. **z measurement:** default = four pixel layers, σ_z = 25 µm; V-stereo adds TIB 1–2 (0.25 mm) and TOB 1–2 (0.5 mm); V-allpixel = all 14 layers at 25 µm, a toy stand-in for all-pixel trackers [9, 10]. **r–z fit:** linear least squares with s from the fitted vertex; $H_{conv,L}$ (z_v, common slope) and $H_{ALP,L}$ (z_v, c_1, c_2); covariance = z resolution + polar scattering (true path lever arms, true initial momentum); $\Delta\chi^2_L = 0$ for pairs without r–z information (each track needs ≥ 2 z hits). **Reference distributions** (regularity checked in advance, v1.6 future direction (vi)): χ^2_1 for $\Delta\chi^2_L$ (Wilks [11]), $½\chi^2_0 + ½\chi^2_1$ for $\Delta\chi^2_T$ (v1.5 [12]), $½\chi^2_1 + ½\chi^2_2$ for $\Delta\chi^2_{3D}$ [13].

3. Validation (independent samples only)

Z1 (zero opening, no brems; 972 of 8000 pairs with r–z information): $\Delta\chi^2_L$ median 0.428 [0.30–0.65], 5.04% above 3.84 [3–8%]; $\Delta\chi^2_{3D}$ median 0.820 [0.60–1.15], 8.64% above 3.84 [6–14%] — pass. The expected values are 0.455 and 5% (χ^2_1) and 0.867 and 9.8% ($½\chi^2_1 + ½\chi^2_2$): the boundary-aware references fixed in advance describe the null distributions. **Z2** (10 GeV pairs at r = 30–100 mm): median $\Delta\chi^2_L$ 0.42, 2.7, 10.4, 41.7 for a polar opening of 0, 0.5, 1, 2 mrad — monotonic, pass. **Z3**: z hits agree with $z_v + \cot\theta \cdot s$ to 2×10^{-7} µm — pass.

4. Results

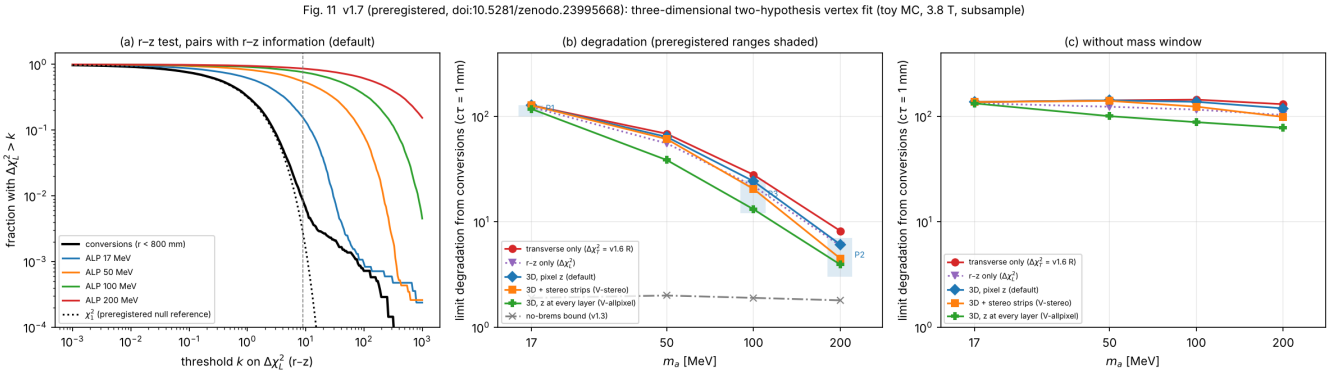


Figure 11. (a) $\Delta\chi^2_L$ survival fractions for conversions and ALP pairs with r–z information (default geometry) and the χ^2_1 reference. (b) Degradation versus mass for the transverse-only, r–z-only and 3D selections and the two variants, with the preregistered ranges. (c) The same without mass window.

selection	17 MeV	50 MeV	100 MeV	200 MeV	no mass window 17 / 200
$\Delta\chi^2_T$ only (= v1.6 config R)	127	68	28	8.2	137 / 130
$\Delta\chi^2_L$ only (pixel z)	122	55	22	5.9	133 / 103
$\Delta\chi^2_{3D}$, pixel z (default)	**127**	**64**	**24**	**6.1**	137 / 119
$\Delta\chi^2_{3D}$, V-stereo	127	61	21	4.5	137 / 99
$\Delta\chi^2_{3D}$, V-allpixel	117	39	13	3.9	133 / 78

Table 1. Limit degradation from conversions (×), cτ = 1 mm, subsample.

geometry	pairs with r–z info: conv. / ALP 17 / 200 MeV	$\Delta\chi^2_L > 9$ (with info): conv. / 17 / 50 / 100 / 200 MeV
pixel z (default)	13.9% / 15.2% / 34.3%	0.84% / 15.5% / 54.8% / 76.3% / 86.5%
V-stereo	86.6% / 78.6% / 91.9%	2.87% / 5.8% / 23.7% / 45.1% / 65.4%
V-allpixel	100% / 100% / 100%	7.44% / 35.8% / 74.4% / 86.0% / 92.3%

Table 2. Mechanism (Q2).

Preregistered predictions vs. outcome

prediction (frozen)	outcome	verdict
P1: 17 MeV in 100–127× (gain at most ≈ 20%)	127.3× (no gain; equal to the baseline quoted as 127×)	confirmed (at the upper edge)
P2: 200 MeV in 3–7×	6.1×	confirmed
P3: 100 MeV in 12–25×	24.2×	confirmed
P4: conversions with r–z info, $\Delta\chi^2_L > 9 \leq 1\%$	0.84%	confirmed
P5: ALP with r–z info, $\Delta\chi^2_L > 9$: 5–25% (17 MeV), ≥ 60% (200 MeV)	15.5%, 86.5%	confirmed
P6: V-stereo improves 200 MeV by ≥ 10%	6.1× → 4.5× (26%)	confirmed
P7: V-allpixel: ≥ 95% with info; 17 MeV ≥ 25% lower; still > 20×	100%; 127.3× → 117.4× (7.8%); 117×	**refuted** (gain part)

5. Deviations from preregistration

(1) Sanity gate 3 failed with the preregistered r–z definition, and the code was fixed. With each r–z hypothesis using the path lengths from its own transverse fit, $\Delta\chi^2_L$ was negative for 2.0% / 2.4% / 12.7% of conversions (default / stereo / allpixel; 1.1–5.5% for signal) against the frozen limit of 0.1%. Diagnosis on conversions: for these pairs the H_{ALP} transverse vertex lies a median 31 mm from the H_{conv} vertex (6 mm for all pairs), i.e. it slides along the nearly parallel tracks — the flat direction diagnosed in v1.5 — so the two r–z fits use inconsistent path lengths and are no longer nested. Fix: both r–z hypotheses use s from the H_{conv} transverse fit (the constrained, v1.4-identical vertex), which makes the r–z test nested; the gate then passed (0.000% negative in all samples and geometries). The fix was chosen on this diagnosis before any degradation was computed; the preregistered-definition values are stored in the outputs. Validations Z1–Z3 use the same definition. **(2)** Code: optional path recording in *minitrack.simulate_track* and stored fit parameters in *fit2h* (defaults unchanged; transverse identity with v1.6 config R 100.000% in all samples). **(3)** Z2 uses a separate longitudinal seed (1808), not specified in the preregistration. No other changes.

6. Interpretation

The r–z view works where the transverse view failed. A polar opening enters the hits at first order and is not faked by bremsstrahlung in the pixel volume (0.84% of conversions), so from 50 MeV on most ALP pairs with r–z information are separated (Table 2). The r–z test alone is even slightly better than the 3D sum, because $\Delta\chi^2_T$ adds the bremsstrahlung-faked transverse openings of v1.5. **Why 17 MeV is not recovered.** In the default geometry only 15% of 17 MeV pairs decay early enough to give r–z information. With z at every layer all pairs have it, yet the gain is only 7.8%: the fraction of conversions with $\Delta\chi^2_L > 9$ rises from 0.84% to 7.4%. Outside the pixel volume the electrons have lost energy to bremsstrahlung and scatter more than the covariance (true initial momentum) assumes, so scattering produces fake polar openings, while the 17 MeV opening (median 0.42 mrad) is comparable to the polar scattering per layer. Because conversions outnumber the signal by orders of magnitude, a few per cent of fakes are enough to keep the veto in place. Within this toy, bremsstrahlung therefore limits the lightest ALPs in the r–z view too, now through the momentum that sets the scattering rather than through the curvature.

7. Limitations

All v1.0–v1.6 limitations apply (barrel only, idealised layers, no pattern recognition, toy material map, crude counting rule, selection optimised on the same samples, truth used for the material list and the scattering momentum, single-Gaussian bremsstrahlung model). The 3D fit is sequential, not joint; one z resolution per layer type; the pair mass is generated independently of the hits; V-allpixel changes only the z coverage, not the material of an all-pixel tracker. Subsampled statistics. Only trends and ratios are meaningful.

8. Future directions

Planned next steps, each under a public preregistration: (i) a **Gaussian-sum filter** [4, 5] (ACTS) with 3D propagation, which follows the momentum lost to bremsstrahlung and thereby the scattering that now fakes polar openings; (ii) an **all-pixel tracker with reduced material**, combining V-allpixel with the v1.6 material scan (the material reduction was the most effective handle there and also reduces the scattering identified here), and a first layer closer to the beam. This is the combination pursued by next-generation tracker concepts — monolithic all-pixel layers with a very low material budget in ALICE 3 [9] and the FCC-ee detectors at about 2 T [10]. Taken together, v1.6 and v1.7 suggest, within this toy, that for the lightest ALPs full z coverage alone is not enough (V-allpixel: 7.8% gain at 17 MeV) and material reduction alone leaves a large loss (v1.6: 35× at a quarter of the material), so a toy configuration of that kind is the natural next test; no statement about the performance of these detectors is made; (iii) a **joint 3D vertex fit** instead of the sequential fit; (iv) the dimuon control study (no bremsstrahlung), which isolates the electron-specific part of these limits; (v) extension to $\mu^+\mu^-$ above 200 MeV and other long-lived-particle signatures, e.g. $B \to K^{(*)}a$ [14]; (vi) calibrated test statistics as applied here (regularity checked in advance; boundary mixtures [12, 13] or pseudo-experiments [15]).

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Links

This record: doi:10.5281/zenodo.24011489 · code:
github.com/wspaik5/ALP_ttbar_Toy_MC_v1.7_Postregistration_3D_Vertex_Fit
Preregistration: doi:10.5281/zenodo.23995668 ·
github.com/wspaik5/ALP_ttbar_Toy_MC_v1.7_Preregistration_3D_Vertex_Fit